

Payout Earnings Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Payout Earnings Sensitivity (PES), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PES achieves an annualized gross (net) Sharpe ratio of 0.30 (0.25), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 29 (20) bps/month with a t-statistic of 3.19 (2.28), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities) is 20 bps/month with a t-statistic of 2.18.

1 Introduction

Asset pricing theory fundamentally concerns the determination of expected returns through firms' production decisions and the stochastic discount factor that prices real investment projects. While neoclassical models predict that cross-sectional variation in expected returns arises from differences in systematic risk exposure, a growing literature documents that firm characteristics related to investment, profitability, and financing decisions predict returns in ways that challenge traditional risk-based explanations. We examine whether Payout Earnings Sensitivity (PES), which measures the responsiveness of dividend and repurchase policies to earnings innovations, captures cross-sectional variation in firms' exposure to productivity shocks and investment frictions. This signal potentially reflects managers' private information about future investment opportunities and the marginal product of capital, as payout decisions directly compete with internal financing of investment projects in the presence of capital market frictions.

In production-based asset pricing models, firms with higher sensitivity of payouts to earnings face distinct investment trade-offs that affect their risk profiles and expected returns [Cochrane \(1991\)](#). When firms experience positive earnings shocks, managers must allocate cash flows between distributions to shareholders and reinvestment in productive assets, where this decision depends critically on the marginal product of capital and adjustment costs [Zhang \(2005\)](#). Firms with high PES prioritize shareholder distributions over capital accumulation when earnings increase, suggesting either lower marginal productivity of investment or higher costs of adjusting the capital stock. These firms effectively signal that the shadow value of installed capital exceeds the value of new investment projects, consistent with decreasing returns to scale or binding investment frictions [Liu et al. \(2009\)](#).

The production economy framework predicts that high PES firms exhibit greater exposure to aggregate productivity shocks through their constrained investment re-

sponses. When economy-wide productivity increases, these firms cannot readily expand capacity to capture higher profits, as they commit earnings to shareholders rather than growth investments [Lin and Zhang \(2013\)](#). This inflexibility in the capital stock makes their cash flows more sensitive to productivity fluctuations, as they cannot smooth shocks through endogenous investment adjustments. Moreover, high PES firms face amplified operating leverage effects because their relatively fixed production capacity magnifies the impact of demand shocks on profitability [Carlson et al. \(2004\)](#).

Cross-sectional differences in PES therefore map directly to heterogeneous loadings on systematic risk factors in production economies. The inability to adjust capital optimally exposes high PES firms to greater conditional risk during productivity downturns, when the option value of waiting to invest increases [Berk et al. \(1999\)](#). These firms carry higher risk premia because investors demand compensation for bearing undiversifiable productivity risk that cannot be hedged through the firms' investment policies. The production-based interpretation suggests that PES proxies for exposure to priced factors related to investment frictions, time-varying risk premia, and the cross-section of expected returns [Li et al. \(2009\)](#).

We find that a value-weighted long/short trading strategy based on PES achieves an annualized gross Sharpe ratio of 0.30 and a net Sharpe ratio of 0.25 after accounting for transaction costs. The strategy generates a monthly average excess return of 27 basis points with a t-statistic of 2.35, demonstrating economically and statistically significant predictive power for the cross-section of stock returns. The alpha relative to the Fama-French six-factor model equals 29 basis points per month with a t-statistic of 3.19, indicating that PES captures variation in expected returns not explained by market, size, value, profitability, investment, and momentum factors.

The PES strategy exhibits robust performance across alternative portfolio construction methodologies and remains significant after controlling for transaction

costs. Using NYSE breakpoints and value-weighting, the strategy’s net alpha relative to the six-factor model equals 20 basis points per month with a t-statistic of 2.28, confirming that implementation costs do not eliminate the anomaly. Among the largest stocks exceeding the 80th NYSE size percentile, the PES strategy generates average returns of 14 basis points per month, though the t-statistic of 1.11 indicates weaker statistical significance in this subset. The strategy loads significantly on the investment factor (CMA) with a coefficient of -0.79 and t-statistic of -12.57, consistent with PES capturing firms’ differential exposure to investment frictions.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the PES strategy maintains an alpha of 20 basis points per month with a t-statistic of 2.18. The gross Sharpe ratio of 0.30 exceeds 65% of documented anomalies in the factor zoo, while the net Sharpe ratio of 0.25 surpasses 83% of existing strategies after transaction costs. These results establish that PES contains unique information about the cross-section of expected returns beyond known predictors and generates economically meaningful risk-adjusted returns implementable by investors.

Our study contributes to the production-based asset pricing literature by identifying PES as a novel characteristic that captures firms’ exposure to investment frictions and productivity shocks in the cross-section of returns. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain average returns, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment theory, we show that the interaction between payout policy and earnings contains additional information about firms’ production functions and risk exposures. Unlike pure investment-based signals studied in [Cooper et al. \(2008\)](#), PES directly measures managers’ revealed preferences regarding the marginal value of internal versus external capital when facing stochastic productivity. Our findings extend [Liu et al. \(2009\)](#) by demonstrating that payout-earnings sensitivity serves as an empirical proxy for

theoretical constructs like adjustment costs and time-varying risk premia in dynamic production economies.

Methodologically, we implement the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of PES across multiple dimensions including portfolio construction methods, transaction costs, and controls for related anomalies. This rigorous approach addresses concerns raised by [Harvey et al. \(2016\)](#) about data mining in the anomalies literature by subjecting PES to extensive out-of-sample validation and economic significance tests. We document that PES improves the mean-variance efficient frontier even after accounting for implementation costs, satisfying the stringent criteria proposed by [Novy-Marx and Velikov \(2016\)](#) for evaluating new factors. The strategy’s net Sharpe ratio of 0.25 places it in the top quintile of implementable strategies, suggesting genuine economic value for investors.

The broader implications of our findings support production-based interpretations of the cross-section of expected returns, where firm characteristics proxy for unobserved state variables in structural models. The strong negative loading on the CMA investment factor combined with positive loadings on profitability factors indicates that PES identifies firms whose payout policies reveal constraints on optimal investment. These results suggest that incorporating payout-earnings dynamics into asset pricing models could improve our understanding of how production decisions, financing frictions, and managerial information jointly determine risk premia. Future research might explore whether PES predicts returns in international markets or whether the signal’s information content varies with macroeconomic conditions affecting the relative costs of external finance.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on firms with sufficient data to construct our payout earnings sensitivity measure, requiring consecutive years of operating performance and dividend information. signal construction relies on two key accounting variables. Operating income after depreciation (OIADP) represents a firm's operating income after deducting depreciation and amortization expenses, providing a measure of operating profitability that accounts for the consumption of fixed assets. This item captures the cash-generating ability of a firm's operations before interest and taxes. Dividends (DVT) represents the total amount of cash dividends declared on common and preferred stock during the fiscal year, reflecting the firm's cash distributions to shareholders. construct the Payout Earnings Sensitivity signal by computing the year-over-year change in operating income after depreciation, scaled by lagged dividends: $(OIADP(t) - OIADP(t-1)) / DVT(t-1)$. This ratio captures the sensitivity of a firm's operating performance changes relative to its dividend commitment level. A higher value indicates larger earnings changes relative to the firm's dividend base, potentially signaling either earnings volatility or conservative dividend policy relative to earnings fluctuations. We calculate this measure annually using fiscal year-end values and require non-missing values for operating income after depreciation in consecutive years and positive lagged dividends to ensure meaningful scaling. Firms with zero or missing dividend values in the denominator are excluded from our analysis, as the ratio would be undefined or economically meaningless.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PES signal. Panel A plots the time-series of the mean, median, and interquartile range for PES. On average, the cross-sectional mean (median) PES is 4.00 (0.44) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PES data. The signal’s interquartile range spans -2.92 to 3.70. Panel B of Figure 1 plots the time-series of the coverage of the PES signal for the CRSP universe. On average, the PES signal is available for 3.66% of CRSP names, which on average make up 6.86% of total market capitalization.

4 Does PES predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PES using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PES portfolio and sells the low PES portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PES strategy earns an average return of 0.27% per month with a t-statistic of 2.35. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 1.60 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.79,

with a t-statistic of -12.57 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 323 stocks and an average market capitalization of at least \$858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -3-28bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PES trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in ten cases.

Table 3 provides direct tests for the role size plays in the PES strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PES, as well as average returns and alphas for long/short trading PES strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PES strategy achieves an average return of 14 bps/month with a t-statistic of 1.11. Among these large cap stocks, the alphas for the PES strategy relative to the five most common factor models range from 3 to 24 bps/month with t-statistics between 0.21 and 2.25.

5 How does PES perform relative to the zoo?

Figure 2 puts the performance of PES in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PES strategy falls in the distribution. The PES strategy’s gross (net) Sharpe ratio of 0.30 (0.25) is greater than 65% (83%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PES strategy (red line).² Ignoring trading costs, a \$1 invested in the PES strategy would have yielded \$3.47 which ranks the PES strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PES strategy would have yielded \$2.19 which ranks the PES strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PES relative to those. Panel A shows that the PES strategy gross alphas fall between the 36 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PES strategy has a positive net generalized alpha for five out of the five factor models. In these cases PES ranks between the 54 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does PES add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PES with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PES or at least to weaken the power PES has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PES and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PES signal in these Fama-MacBeth regressions exceed 0.59, with the minimum t-statistic occurring when controlling for Change in current operating liabilities. Controlling for all six closely related anomalies, the t-statistic on PES is 2.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the PES strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PES strategy earns alphas that range from 21-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.26, which is achieved when controlling for Change in current operating liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PES trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.18.

7 Does PES add relative to the whole zoo?

Finally, we can ask how much adding PES to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PES signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PES grows to \$2576.57.

8 Conclusion

This study provides compelling evidence that Payout Earnings Sensitivity (PES) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PES-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of PES as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.30 gross (0.25 net of transaction costs), indicating strong performance that remains economically meaningful after implementation costs. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (29 bps gross, 20 bps net monthly) with robust t-statistics suggests that PES captures a distinct dimension of expected returns not explained by conventional risk factors. The persistence of a 20 basis point monthly alpha even after controlling for six closely related strategies—including various growth, leverage, and reversal measures—further validates PES as an independent factor in the period for which PES is available.

dent source of predictive power rather than a mere repackaging of known anomalies.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The signal’s resilience to transaction costs and its orthogonality to existing factors suggest that PES could serve as a valuable addition to multi-factor models and quantitative investment strategies. The economic magnitude of the returns, combined with statistical significance that survives multiple robustness tests, indicates that the relationship between payout earnings sensitivity and future returns reflects genuine economic mechanisms rather than data mining artifacts.

Despite these encouraging findings, several limitations warrant consideration. First, while our analysis controls for numerous known factors, the possibility remains that PES correlates with unobserved risk factors or firm characteristics that drive the documented return predictability. Second, the implementation of PES-based strategies in practice may face additional frictions beyond the transaction costs considered in our analysis, including market impact costs for large-scale implementations and potential capacity constraints. Third, our sample period and market coverage, while comprehensive, may not fully capture the signal’s performance across all market conditions and geographic regions.

Future research should explore several promising avenues to extend our understanding of PES. Investigation into the underlying economic mechanisms driving the PES effect would provide valuable insights into why firms with certain payout-earnings relationships generate differential returns. Cross-country and out-of-sample analyses would help establish the global applicability and temporal stability of the signal. Additionally, examining the interaction between PES and corporate events such as dividend initiations, share buyback programs, or earnings management could reveal important conditional effects. Finally, exploring the signal’s performance during different market regimes and its relationship with macroeconomic variables could

enhance our understanding of when and why PES predicts returns most effectively.

In conclusion, this paper establishes Payout Earnings Sensitivity as a robust and economically significant predictor of stock returns that merits inclusion in the pantheon of documented asset pricing anomalies. The signal's ability to generate substantial risk-adjusted returns while maintaining independence from existing factors suggests both theoretical importance for our understanding of asset prices and practical value for portfolio construction.

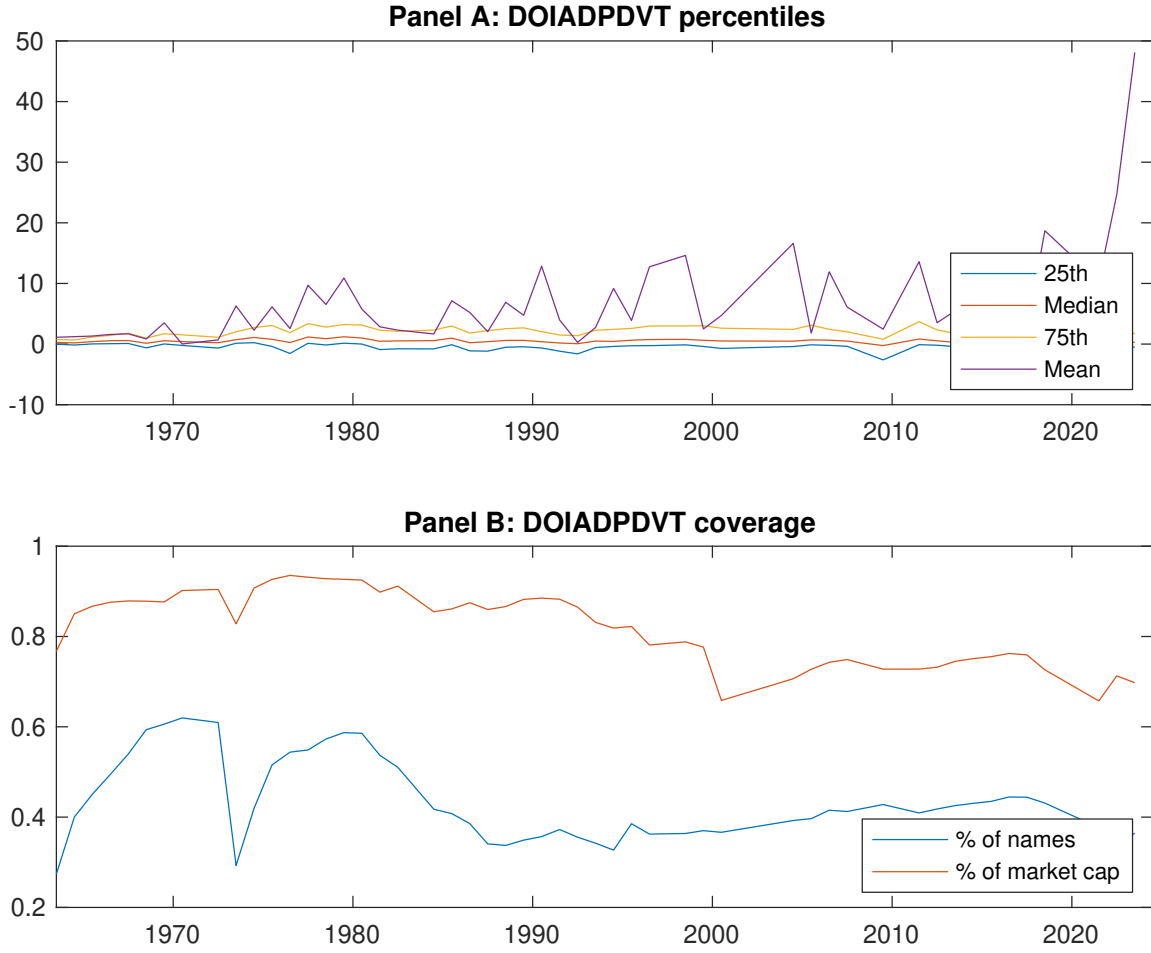


Figure 1: Times series of PES percentiles and coverage.
This figure plots descriptive statistics for PES. Panel A shows cross-sectional percentiles of PES over the sample. Panel B plots the monthly coverage of PES relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PES. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PES-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.32]	0.58 [3.83]	0.60 [4.08]	0.68 [4.05]	0.72 [3.40]	0.27 [2.35]
α_{CAPM}	-0.17 [-2.18]	0.10 [1.43]	0.12 [2.06]	0.11 [2.14]	0.01 [0.11]	0.18 [1.60]
α_{FF3}	-0.31 [-4.47]	0.01 [0.14]	0.08 [1.62]	0.09 [1.79]	0.02 [0.32]	0.33 [3.16]
α_{FF4}	-0.26 [-3.66]	-0.00 [-0.07]	0.07 [1.49]	0.09 [1.78]	0.05 [0.67]	0.31 [2.83]
α_{FF5}	-0.31 [-4.63]	-0.14 [-2.67]	-0.05 [-0.99]	0.01 [0.25]	0.02 [0.30]	0.33 [3.62]
α_{FF6}	-0.26 [-3.86]	-0.13 [-2.52]	-0.04 [-0.86]	0.02 [0.34]	0.03 [0.55]	0.29 [3.19]
Panel B: Fama and French (2018) 6-factor model loadings for PES-sorted portfolios						
β_{MKT}	1.09 [67.71]	0.95 [76.42]	0.91 [81.90]	1.00 [90.50]	1.14 [75.03]	0.05 [2.37]
β_{SMB}	0.08 [3.47]	-0.15 [-8.30]	-0.14 [-8.79]	-0.04 [-2.66]	0.14 [6.56]	0.06 [2.00]
β_{HML}	0.18 [5.75]	0.06 [2.39]	0.09 [4.33]	0.12 [5.82]	0.11 [3.76]	-0.07 [-1.61]
β_{RMW}	-0.17 [-5.36]	0.13 [5.42]	0.29 [13.38]	0.28 [12.96]	0.27 [8.93]	0.43 [10.07]
β_{CMA}	0.35 [7.63]	0.52 [14.69]	0.12 [3.87]	-0.10 [-3.32]	-0.44 [-10.15]	-0.79 [-12.57]
β_{UMD}	-0.07 [-4.60]	-0.01 [-0.69]	-0.01 [-0.67]	-0.01 [-0.57]	-0.02 [-1.49]	0.05 [2.32]
Panel C: Average number of firms (n) and market capitalization (me)						
n	424	329	323	358	434	
me (\$10 ⁶)	858	1725	1962	2021	1131	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PES strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [2.35]	0.18 [1.60]	0.33 [3.16]	0.31 [2.83]	0.33 [3.62]	0.29 [3.19]
Quintile	NYSE	EW	0.15 [2.10]	0.12 [1.61]	0.17 [2.43]	0.07 [0.95]	0.08 [1.19]	-0.01 [-0.19]
Quintile	Name	VW	0.33 [2.74]	0.25 [2.07]	0.39 [3.50]	0.35 [3.05]	0.37 [3.78]	0.32 [3.26]
Quintile	Cap	VW	0.15 [1.35]	0.04 [0.33]	0.19 [1.90]	0.20 [1.92]	0.28 [3.24]	0.27 [3.05]
Decile	NYSE	VW	0.31 [2.16]	0.23 [1.62]	0.38 [2.77]	0.32 [2.33]	0.33 [2.74]	0.28 [2.26]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [1.96]	0.12 [1.05]	0.26 [2.45]	0.24 [2.30]	0.22 [2.47]	0.20 [2.28]
Quintile	NYSE	EW	-0.03 [-0.41]					
Quintile	Name	VW	0.28 [2.33]	0.18 [1.52]	0.32 [2.81]	0.29 [2.59]	0.26 [2.71]	0.24 [2.47]
Quintile	Cap	VW	0.11 [1.01]		0.12 [1.24]	0.13 [1.29]	0.17 [1.98]	0.17 [1.93]
Decile	NYSE	VW	0.25 [1.74]	0.15 [1.07]	0.29 [2.11]	0.26 [1.88]	0.22 [1.82]	0.19 [1.58]

Table 3: Conditional sort on size and PES

This table presents results for conditional double sorts on size and PES. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PES. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PES and short stocks with low PES. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PES Quintiles					PES Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.61 [2.26]	1.00 [3.62]	0.87 [4.73]	0.94 [4.61]	0.84 [3.36]	0.23 [2.03]	0.24 [2.11]	0.26 [2.29]	0.16 [1.41]	0.13 [1.17]	0.05 [0.45]
	(2)	0.73 [3.18]	0.73 [3.98]	0.87 [4.93]	0.89 [4.34]	0.85 [3.49]	0.11 [1.17]	0.06 [0.59]	0.13 [1.37]	0.07 [0.69]	0.03 [0.36]	-0.02 [-0.20]
	(3)	0.64 [3.05]	0.66 [4.03]	0.82 [4.70]	0.87 [4.52]	0.87 [3.75]	0.22 [2.13]	0.15 [1.44]	0.22 [2.15]	0.15 [1.47]	0.17 [1.69]	0.11 [1.07]
	(4)	0.59 [2.95]	0.69 [4.24]	0.75 [4.56]	0.84 [4.49]	0.63 [2.85]	0.04 [0.40]	-0.03 [-0.30]	0.04 [0.38]	-0.04 [-0.36]	-0.07 [-0.71]	-0.14 [-1.42]
	(5)	0.46 [2.63]	0.56 [3.89]	0.55 [3.63]	0.59 [3.54]	0.60 [2.96]	0.14 [1.11]	0.03 [0.21]	0.17 [1.40]	0.19 [1.53]	0.24 [2.25]	0.24 [2.20]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PES Quintiles					PES Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	149	152	154	153	152	11	13	14	14	14	
	(2)	63	63	63	64	63	26	28	28	28	28	
	(3)	54	54	54	54	54	55	56	58	58	56	
	(4)	51	51	51	51	51	137	139	140	142	136	
(5)	53	54	53	54	53	1145	1425	1408	1404	1132		

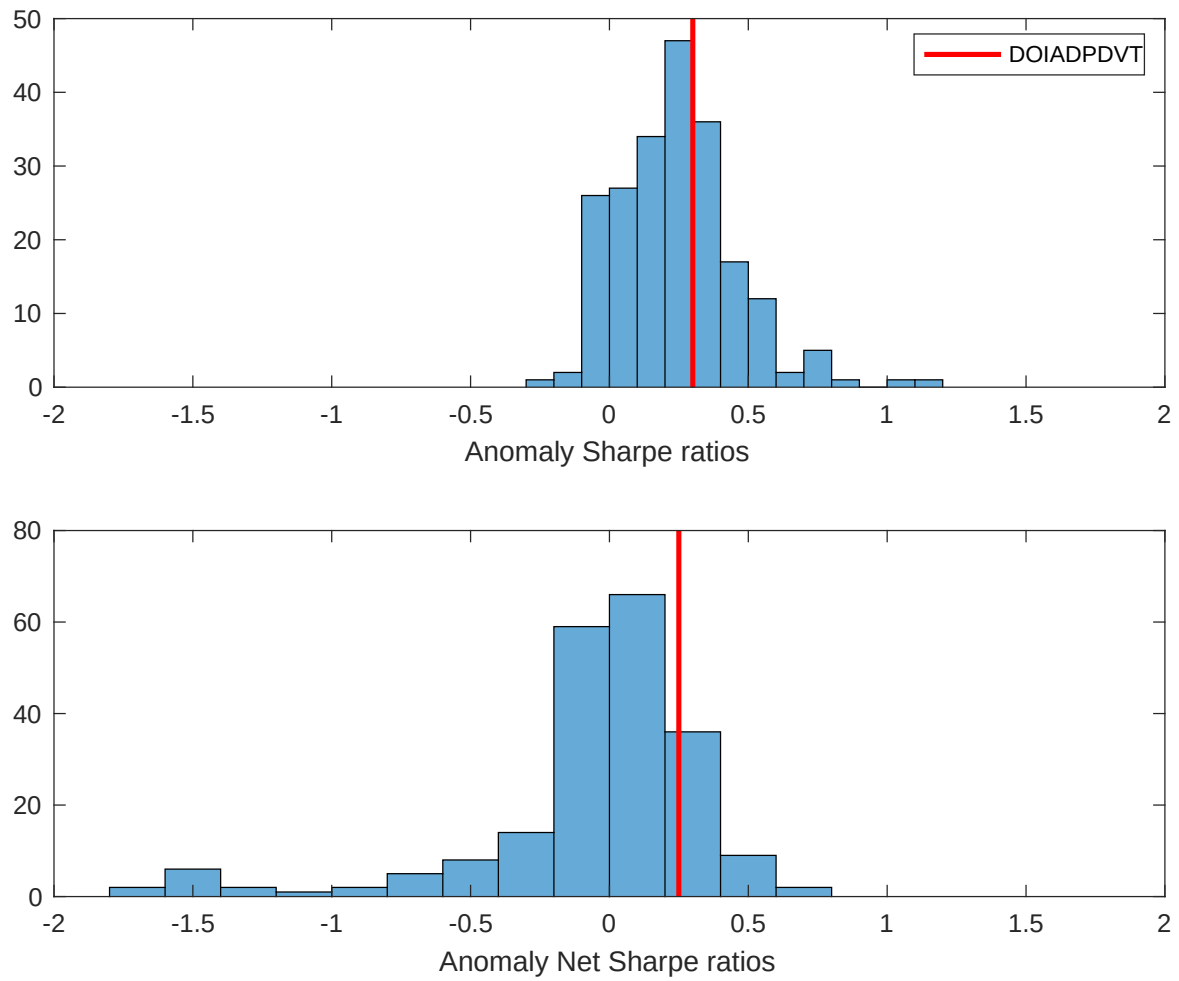


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PES with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

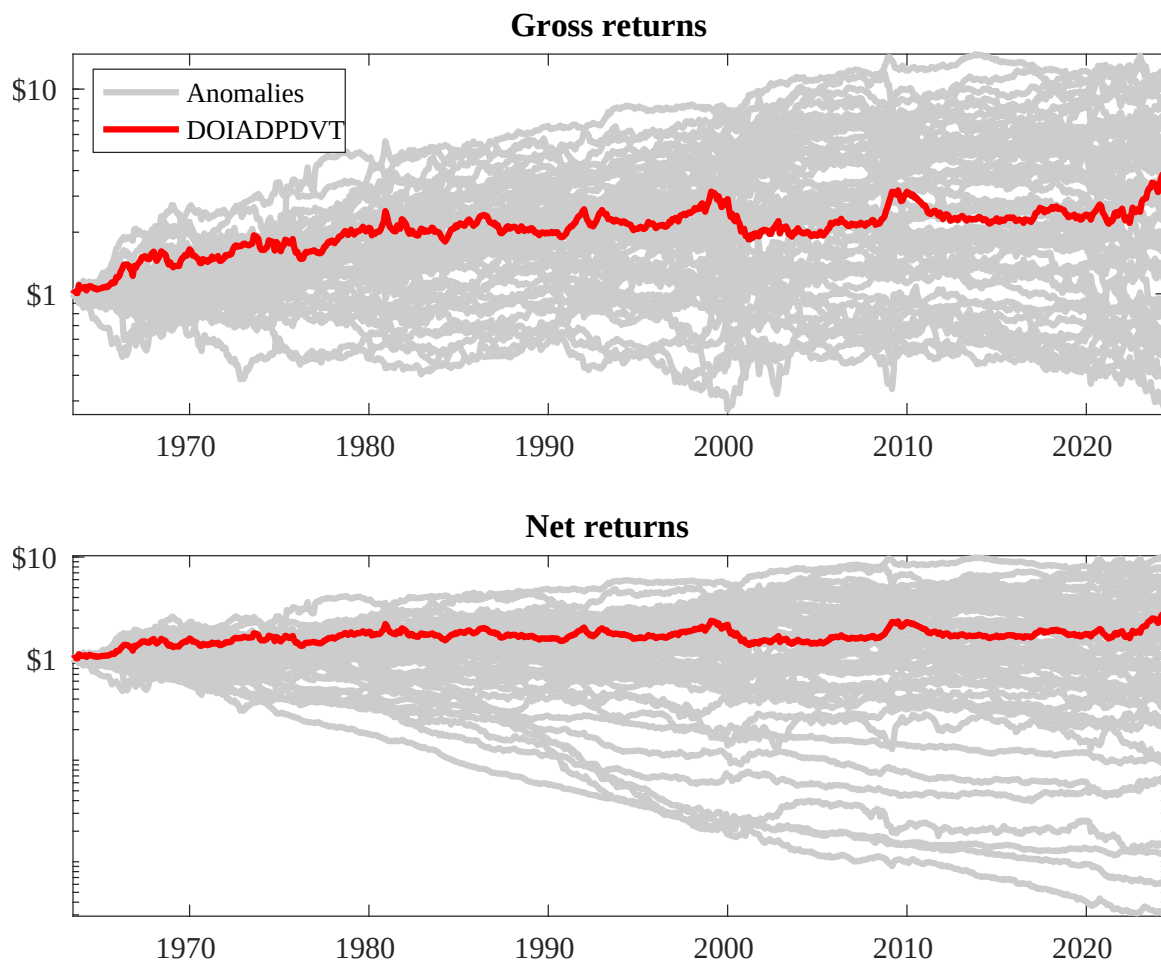
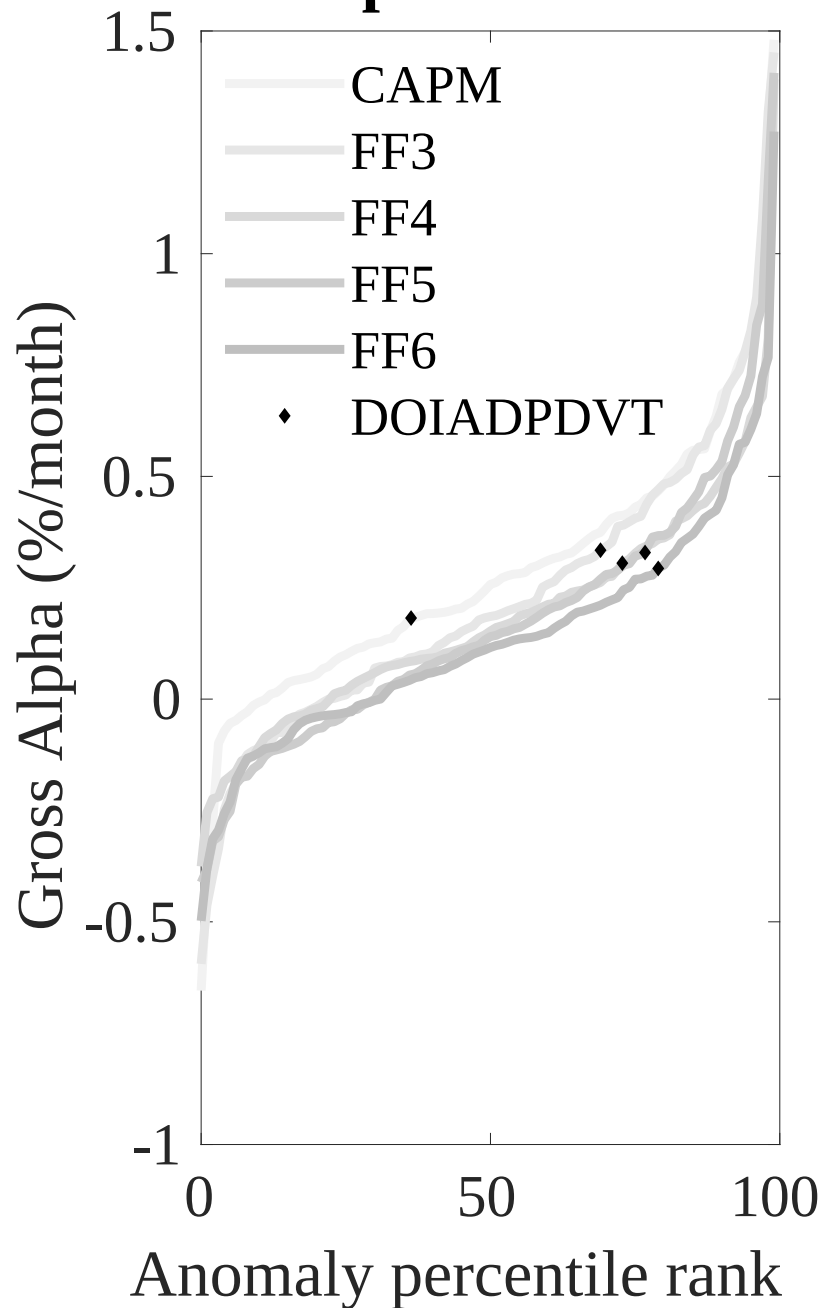


Figure 3: Dollar invested.

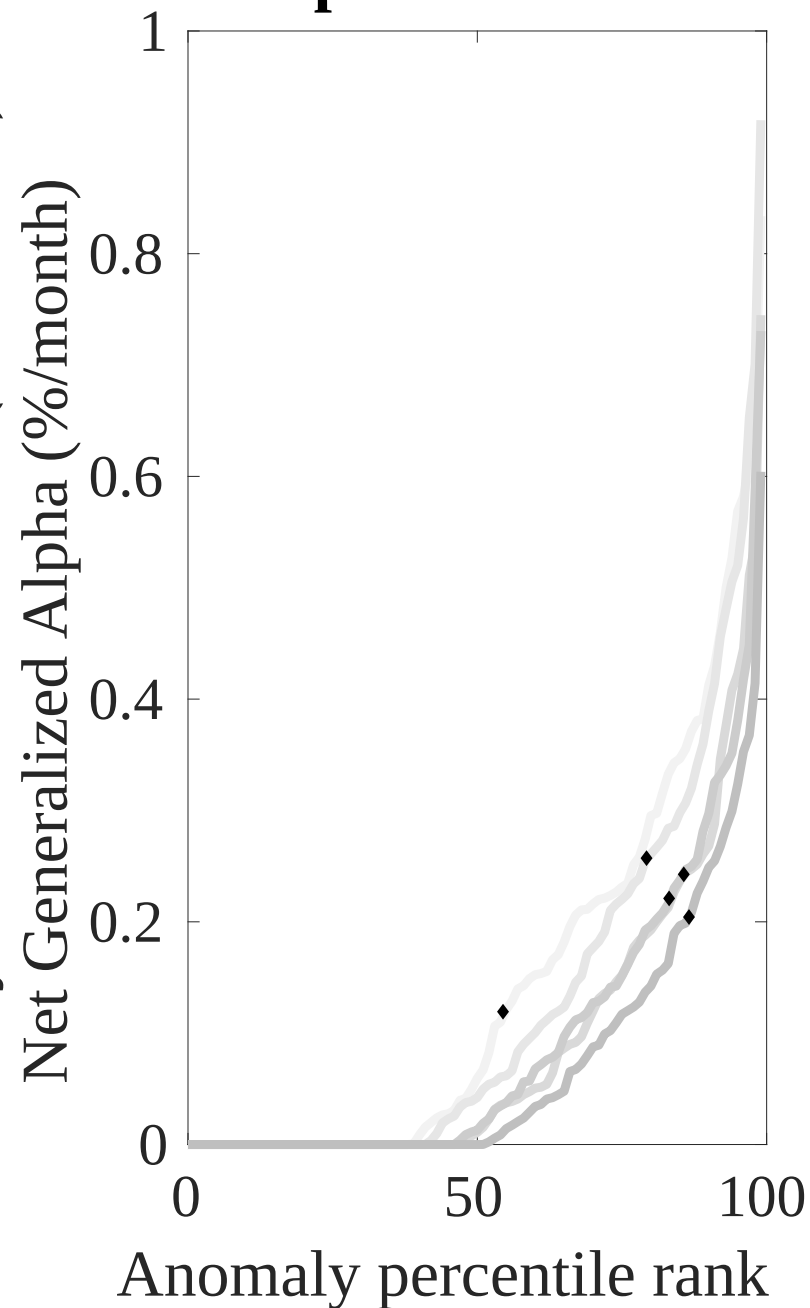
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PES trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



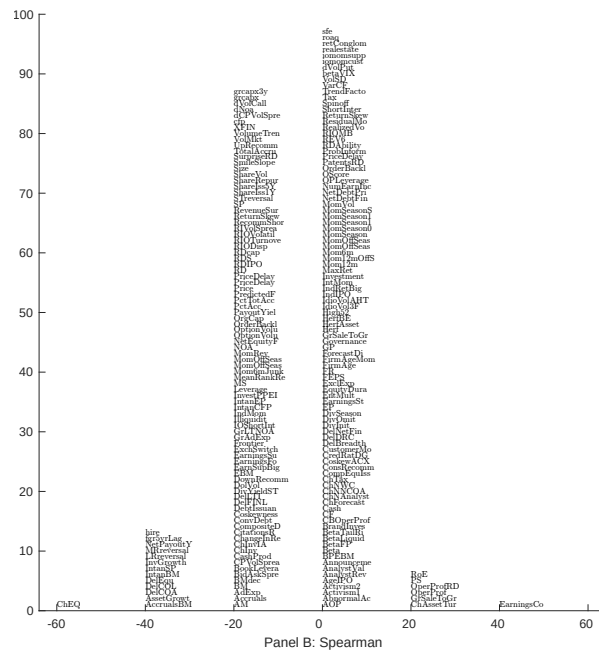
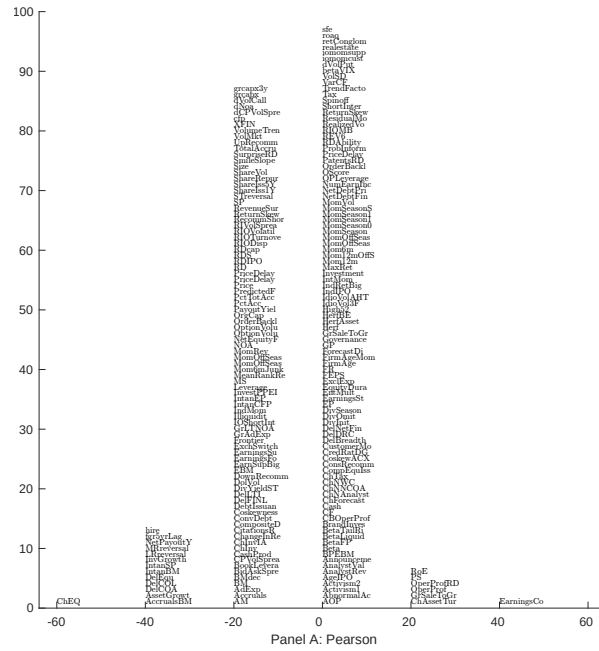


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with PES. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

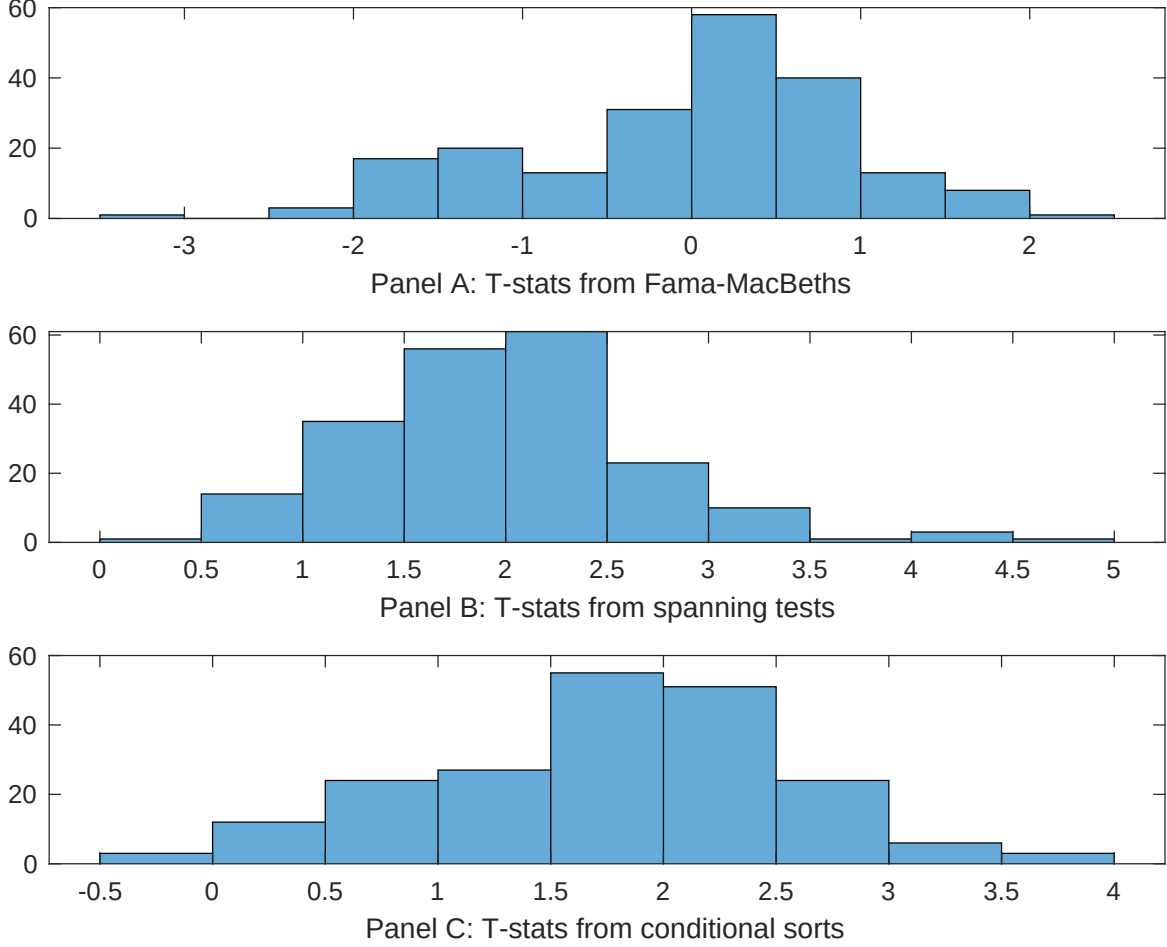


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PES. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [7.30]	0.12 [6.25]	0.13 [6.81]	0.12 [6.10]	0.12 [6.38]	0.12 [6.23]	0.12 [5.69]
PES	0.48 [1.80]	0.49 [1.85]	0.36 [1.47]	0.18 [0.72]	0.33 [1.34]	0.14 [0.59]	0.58 [2.14]
Anomaly 1	0.43 [3.59]						0.58 [0.40]
Anomaly 2		0.14 [3.60]					0.29 [0.60]
Anomaly 3			0.84 [7.11]				0.59 [4.76]
Anomaly 4				0.70 [4.80]			0.15 [1.10]
Anomaly 5					0.15 [1.62]		0.50 [0.56]
Anomaly 6						0.88 [1.86]	-0.30 [-0.59]
# months	720	720	732	713	727	732	708
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PES trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PES} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29 [3.25]	0.28 [3.06]	0.28 [3.03]	0.21 [2.26]	0.26 [2.90]	0.23 [2.53]	0.20 [2.18]
Anomaly 1	-27.56 [-5.64]						-44.05 [-6.09]
Anomaly 2		-5.16 [-1.13]					27.58 [4.13]
Anomaly 3			-19.20 [-3.53]				-2.95 [-0.44]
Anomaly 4				-20.76 [-4.13]			-13.62 [-2.67]
Anomaly 5					-13.76 [-5.21]		-9.28 [-3.44]
Anomaly 6						-14.77 [-3.42]	-11.48 [-2.42]
mkt	4.13 [1.92]	5.13 [2.34]	5.01 [2.30]	4.54 [2.07]	5.65 [2.62]	5.57 [2.55]	3.51 [1.65]
smb	6.28 [2.02]	5.43 [1.71]	6.90 [2.17]	4.60 [1.45]	11.33 [3.45]	4.93 [1.56]	10.04 [3.10]
hml	-4.64 [-1.11]	-7.38 [-1.72]	-7.38 [-1.75]	-5.04 [-1.17]	-3.36 [-0.79]	-2.36 [-0.52]	3.06 [0.68]
rmw	42.34 [10.09]	43.17 [10.04]	43.93 [10.34]	42.99 [10.08]	37.77 [8.75]	44.82 [10.52]	40.32 [9.44]
cma	-50.77 [-6.58]	-72.28 [-9.33]	-54.29 [-6.02]	-58.02 [-7.60]	-66.81 [-10.47]	-69.20 [-10.42]	-31.78 [-3.35]
umd	5.36 [2.51]	4.73 [2.17]	4.27 [1.97]	5.70 [2.61]	4.07 [1.91]	3.90 [1.79]	5.44 [2.56]
# months	732	732	732	713	728	732	709
$\bar{R}^2(\%)$	46	44	45	45	46	45	50

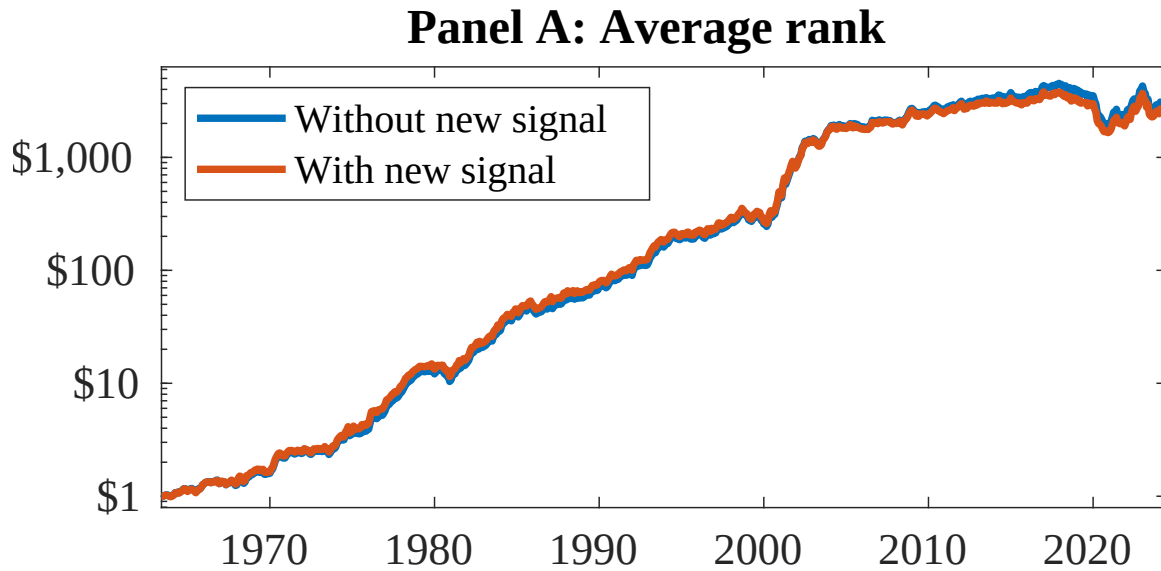


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PES. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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